

Medical Journey Log: A Family Care Tracker
Alex Frear
CST-451 Milestone 4: Development (Coding Phase)
Grand Canyon University
Instructor: Professor Scott Webster
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Screencast Video 1 (Code Review) Link: [Code Review Screencast](#)

Screencast Video 2 (Application Running) Link: [Application Running Screencast](#)

Github Link: [CST - 451 - Milestone 4](#)

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Abstract

The Medical Journey Log is a web-based application designed to help caregivers document and manage medical details for their children. This milestone demonstrates key features of the application, including the ability to create child profiles, log symptoms through multiple entry points, and view a detailed history for each child. Built with ASP.NET Core Razor Pages and MySQL, the system offers a user-friendly interface and a solid foundation for further development. The project was developed and tested on macOS using Visual Studio and MySQL Workbench in a local environment.

Implementation Plan

Use Case/User Story

As a caregiver, I want to create a child profile and log symptoms so that I can track my child's medical history in one place.

Development Tasks – Completed for Milestone 4

Task	Description	Estimate (hrs)	Actual (hrs)	% Complete
1	Set up MySQL database and build EF models for Child and Symptom	3	3	100%
2	Create Razor Page to add a new child	2	2	100%
3	Create LogSymptom page accessible from navbar	2	2	100%
4	Add ability to pass childId to LogSymptom page	2	2	100%
5	Build child details page showing name, DOB, and symptoms	3	3	100%
6	Populate dropdown of children for symptom logging	1	1	100%
7	Test core features manually in browser (child creation, symptom logging)	2	2	100%
8	Prepare .zip and .sql files for submission	1	1	100%

Upcoming Tasks – Planned for Future Milestones

Task	Description	Estimate (hrs)	Status
9	Implement edit/delete functionality for child and symptom entries	4	Planned
10	Create caregiver login system using ASP.NET Identity	5	Planned
11	Support multiple caregivers and children (many-to-many relationship)	5	Planned
12	Add report generation (printable or exportable summaries)	3	Planned
13	Implement responsive design improvements for mobile use	2	Planned
14	Apply advanced validation and form feedback for UX	2	Planned
15	Add navigation improvements and filtering/search options	2	Planned

Summary

- Percent of User Stories complete for this iteration: 100%
- Percent of User Stories complete for entire project: approximately 40%
- Future milestones will expand on this foundation by introducing edit/delete features, user authentication, multi-user support, and report generation.

Traceability Matrix

Functional Requirement	Design Location	Code Implementation	Test Case Reference
Create a new child profile	Create.cshtml wireframe, Milestone 2	Pages/Create.cshtml and Create.cshtml.cs	Test Case TC-001
Log a symptom from navbar	LogSymptom.cshtml wireframe, Milestone 2	Pages/LogSymptom.cshtml and LogSymptom.cshtml.cs	Test Case TC-002
Log a symptom from child detail page	ChildDetails.cshtml link + LogSymptom mockup	Pages/ChildDetails.cshtml and LogSymptom.cshtml.cs with childId parameter	Test Case TC-003
View child details with symptoms listed	ChildDetails.cshtml wireframe, Milestone 2	Pages/ChildDetails.cshtml and ChildDetails.cshtml.cs	Test Case TC-004
Store and retrieve data using MySQL	Architecture Plan, Database Schema (ERD)	ApplicationDbContext.cs, MySQL Workbench, appsettings.json	Verified with live data

Test Plan and Test Cases

Test Objective

To verify that core application features (child creation, symptom logging, and viewing child details) function correctly according to user stories and requirements.

Test Case TC-001 – Create New Child Profile

Field	Value
Module	Child Creation
Test Objective	Verify that a caregiver can successfully create a new child profile
Test Steps	1. Navigate to the "Create Child" page 2. Enter valid child name and date of birth 3. Click "Save"
Test Data	Name: "Test Child 1" DOB: 01/01/2020
Expected Result	New child is saved and user is redirected to home or confirmation page
Actual Result	Passed – child is created and visible in the database
Pass/Fail	Pass

Test Case TC-002 – Log Symptom from Navigation Bar

Field	Value
Module	Symptom Logging
Test Objective	Verify that a caregiver can log a symptom using the main navigation option
Test Steps	1. Click "Log Symptom" in the navbar 2. Select a child from the dropdown 3. Enter a symptom name and description 4. Click "Save"
Test Data	Child: Test Child 1 Symptom: "Headache" Description: "Mild but persistent"
Expected Result	Symptom is saved and tied to the selected child
Actual Result	Passed – symptom appears in database and child detail page
Pass/Fail	Pass

Test Case TC-003 – Log Symptom from Child Detail Page

Field	Value
Module	Symptom Logging with Child Context

Test Objective	Verify that a symptom can be logged directly from a specific child's detail page
Test Steps	1. View details for Test Child 1 2. Click "Log Symptom" button 3. Enter symptom data and submit
Test Data	Symptom: "Fever" Description: "Slightly elevated temperature"
Expected Result	Symptom is saved and auto-associated with Test Child 1
Actual Result	Passed – symptom correctly saved and displayed on detail page
Pass/Fail	Pass

Test Case TC-004 – View Child Detail Page with Symptoms

Field	Value
Module	Child Details Display
Test Objective	Verify that a child's detail page displays correct information and logged symptoms
Test Steps	1. Navigate to child detail page 2. Observe displayed symptoms
Test Data	N/A (assumes previously logged symptoms)
Expected Result	Child name, DOB, and symptom list are visible with correct timestamps and descriptions
Actual Result	Passed – symptoms displayed as expected
Pass/Fail	Pass

Source Code Listing

The following is a brief overview of the key files and classes included in the Medical Journey Log application.

File / Class	Description
Program.cs	Entry point of the application; configures and runs the web host
Startup.cs	Registers services and configures the HTTP request pipeline, including routing and database context
appsettings.json	Contains database connection string and environment configuration values
Data/ApplicationDbContext.cs	Entity Framework database context; manages Child and Symptom models
Models/Child.cs	Defines the Child entity with properties like Name and DateOfBirth
Models/Symptom.cs	Defines the Symptom entity with properties like Name, Description, DateLogged, and foreign key to Child
Pages/Create.cshtml / .cs	Razor Page for creating new child profiles
Pages/LogSymptom.cshtml / .cs	Razor Page for logging symptoms with optional child pre-selection
Pages/ChildDetails.cshtml / .cs	Displays child information and a list of their symptoms
Pages/Index.cshtml / .cs	Home page showing navigation and optional summary info (placeholder for future updates)

All files are written in C# using ASP.NET Core Razor Pages and follow standard MVC separation within Razor Page conventions. The application connects to a MySQL database using Entity Framework Core.